

Altos

A Game-Ready Asset by [OccaSoftware](#)

Need Support?

If you run into any issues or have any questions, please don't hesitate to contact me by email, on Twitter, or on Discord.

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Introduction

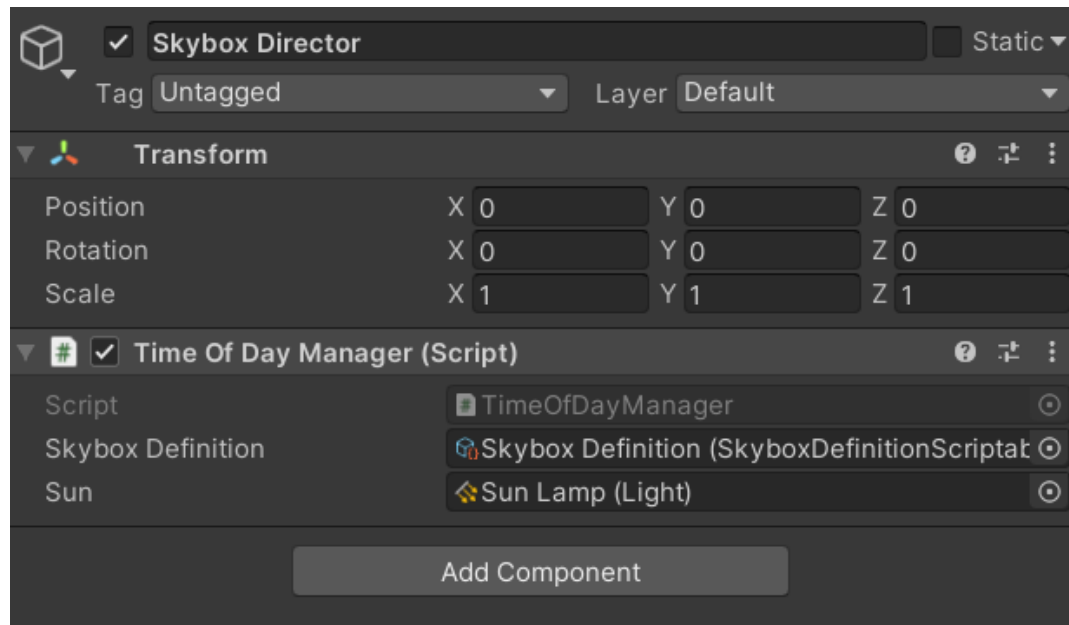
Easily add beautiful dynamic skies, volumetric clouds, and a 24-hour day/night cycle to your game.

Designed for Unity 2021.3 Universal Render Pipeline (URP).

Features

1. Dynamic, Customizable Skybox
2. Time of Day Manager enables massive open world games with realistic day/night cycles
3. Skybox, High Altitude, and Low Altitude Volumetric Clouds
4. Realistic night sky rendering with stars and moon
5. Depth fog synchronized to the scene's lighting and atmospheric
6. Highly customizable and easy to use

First Setup - Skybox



1. Import the Altos package.
2. I strongly recommend exploring the sample scene included in the project.
3. Click on the + button near the top of your Hierarchy window. Click Skybox Director. This is a one-click setup that creates all the needed data and configurations for the Skybox to work. You will need to do additional setup to assign the Moon and Stars objects to the correct layers.
4. Open the Skybox Definition object from your TimeOfDayManager to configure the Skybox and fog configuration that is applied in your scene.

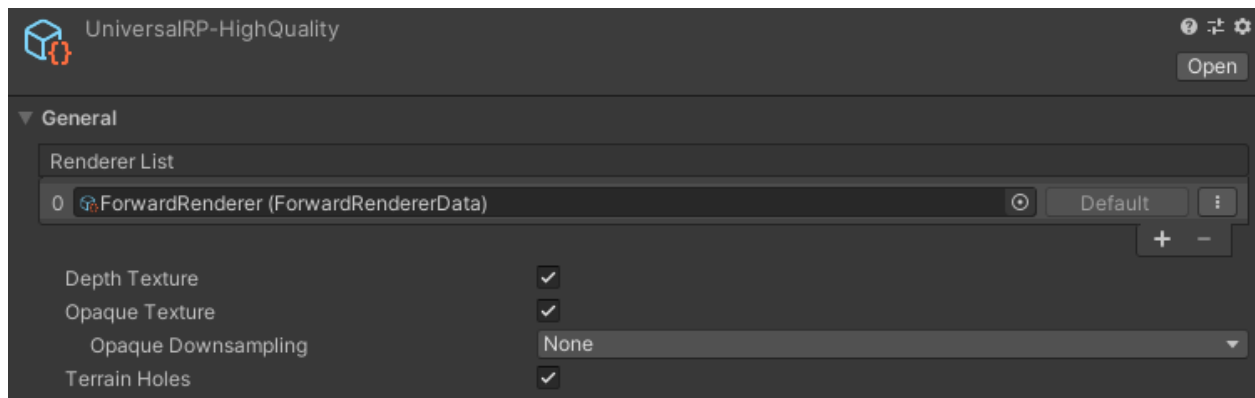
Important Note:

The one-click setup requires you to reload the scene. To do so, enter play mode, and then exit play mode. The Skybox should now be correctly configured.

(I am working to simplify this initial setup.)

First Setup - Volumetric Clouds

1. Open your Forward Renderer in "Assets/Settings/...", click "Add Renderer Feature", and select "Skybox Clouds Feature".
2. Configure your Volumetric Clouds from the Clouds game object under the Skybox Director.
3. Volumetric Clouds requires your Rendering Asset to generate Opaque and Depth Textures. Ensure that you generate these textures from your UniversalRP-[High/Medium/Low]Quality assets.

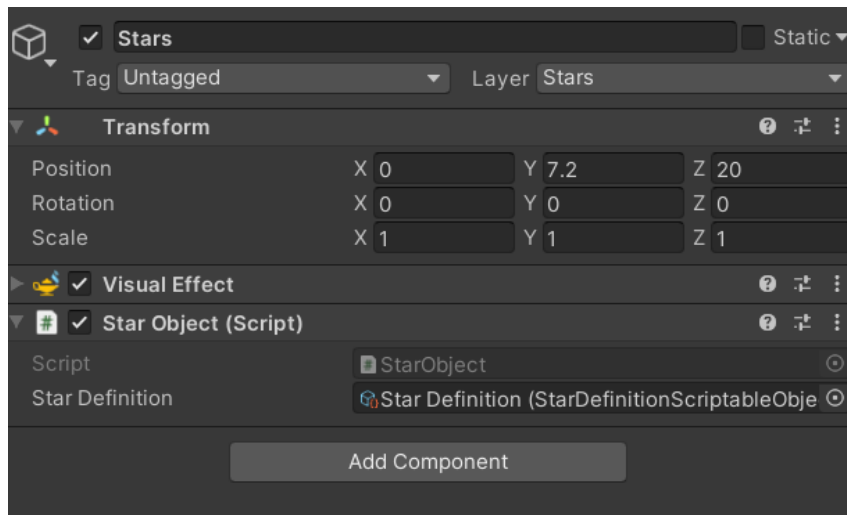


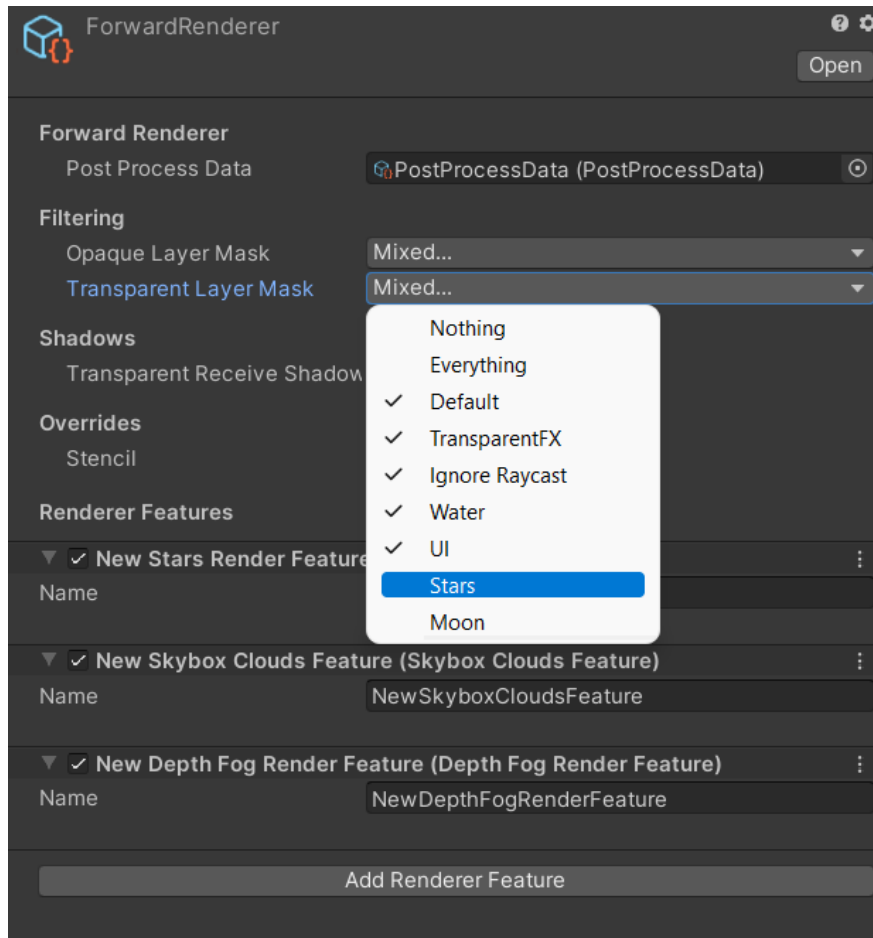
First Setup - Stars and Moon

Stars

Stars can render behind Volumetric Clouds without impacting other Transparent Particles. This requires 2 simple steps.

1. On your Stars Game Object, create and set a new Layer to be used exclusively with the Stars game object. I named mine Stars.
2. On your Forward Renderer, open the Transparent Layer Mask and de-select the newly added layer. Then add the Stars Render Feature.

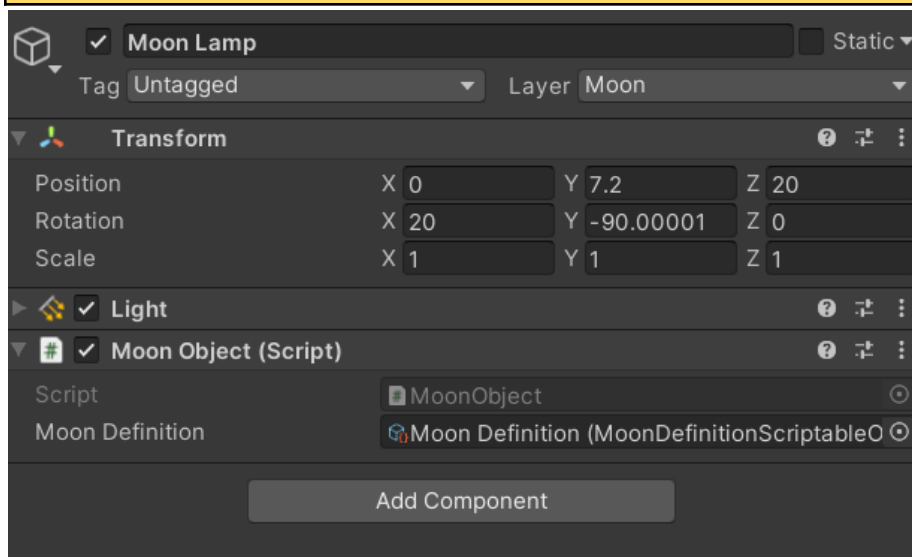




Moon

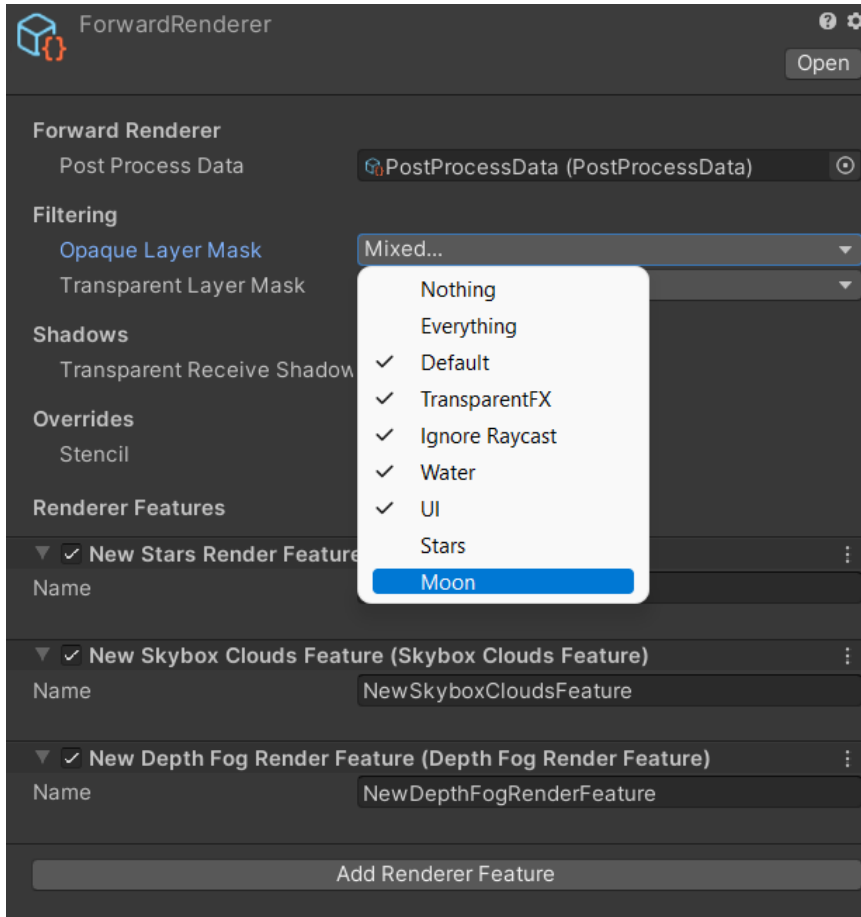
The moon can also render behind Volumetric Clouds without impacting other Transparent Particles. This requires 2 easy steps.

1. On your Moon Game Object, create and set a new Layer to be used exclusively with the Moon game object. I named mine Moon.
2. On your Forward Renderer, pop open the Opaque Layer Mask and de-select the newly added layer. Then add the Stars Render Feature if you have not already added it for the Stars. You should only have one Stars Render Feature present in your ForwardRenderer.

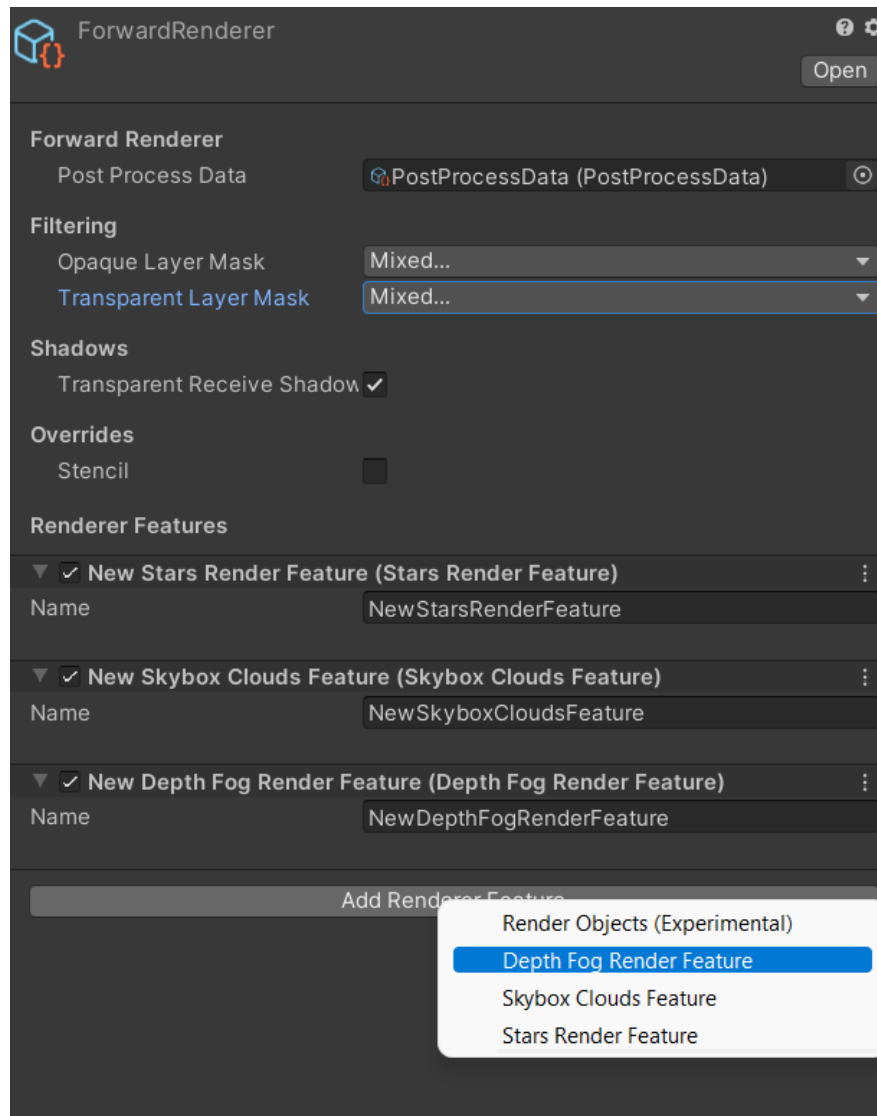


Handy Feature

We've included a lightweight helper class, `AttenuateLightBelowHorizon.cs`, so that your moon can automatically adjust its brightness as it goes below the horizon.



First Setup - Depth Fog



1. In your ForwardRenderer, click "Add Renderer Feature".
2. Select "Depth Fog Render Feature"

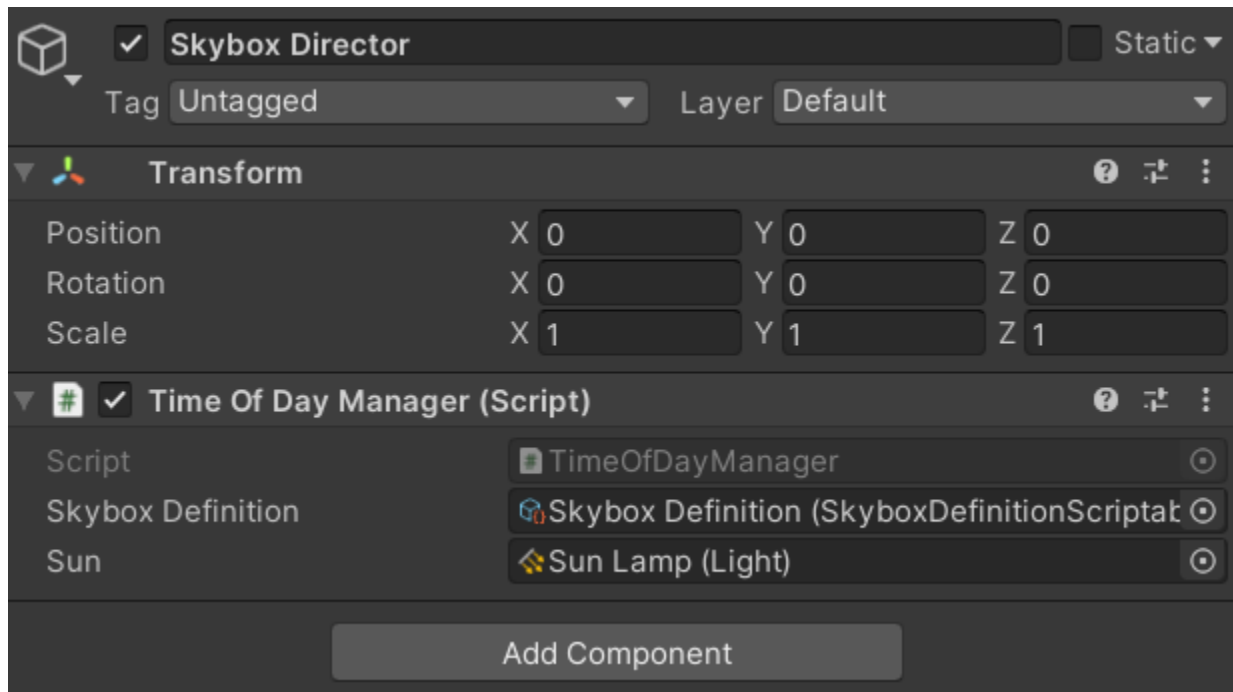
Skybox Definitions

A Skybox Definition represents a pre-made configuration asset for the Times of Day, Sky, Cloud, Star, Fog, and other settings. You can save these assets, use them across scenes and have changes in one asset propagate to all scenes, or swap out to different Skybox Definitions while in a given scene.

We have distinct Definition Objects for your Time of Day Manager, Stars, Moon, and Clouds configurations. This enables you to easily save and swap your favorite configurations. Animated transitions are not supported at this time.

You can create a new asset for each configuration by right clicking in your Project window and selecting Skies and choosing the appropriate sub-selection.

Example Configuration





☒ **Sun Lamp**

☐ Static

Tag

Untagged

Layer

Default

 Transform

Position

X

0

Y

0

Z

0

Rotation

X

52.5

Y

36

Z

180

Scale

X

1

Y

1

Z

1

 ☒ Light

Type

Directional

Use color temperature mode

☒

Color

Filter

Temperature

4800

Mode

Realtime

Intensity

1.26

Indirect Multiplier

1

Shadow Type

Soft Shadows

Realtime Shadows

Strength

1

Bias

Use Pipeline Settings

Near Plane

0.2

Render Mode

Auto

Culling Mask

Everything

 ☒ Sun Lamp Object (Script)

Script

SunLampObject

Add Component



☒ Moon Lamp

☐ Static

Tag

Untagged

Layer

Moon

▼  Transform

Position

X

0

Y

7.2

Z

20

Rotation

X

20

Y

-90.00001

Z

0

Scale

X

1

Y

1

Z

1

▼  ☒ Light

Type

Directional

Use color temperature mode

☒

Color

Filter

Temperature

10000

Mode

Realtime

Intensity

1

Indirect Multiplier

1

Shadow Type

No Shadows

Render Mode

Auto

Culling Mask

Everything

▼  ☒ Moon Object (Script)

Script

MoonObject

Moon Definition

Moon Definition (MoonDefinitionScriptableO

Add Component



☒ Stars

☐ Static

Tag

Untagged

Layer

Stars



Transform



Position

X

0

Y

7.2

Z

20

Rotation

X

0

Y

0

Z

0

Scale

X

1

Y

1

Z

1



☒ Star Object (Script)



Script

 StarObject

Star Definition

 Star Definition (StarDefinitionScriptableObje



☒ Visual Effect



Add Component



☒ Clouds

☐ Static

Tag

Untagged

Layer

Default

▼ 

Transform

?





Position

X

0

Y

0

Z

0

Rotation

X

0

Y

0

Z

0

Scale

X

1

Y

1

Z

1

▼ 

Volumetric Cloud Volume (Script)

?





Script

 VolumetricCloudVolume



Cloud Data

 Cloud Definition (VolumetricCloudsDefinitior



Add Component



ForwardRenderer



Open

Forward Renderer

Post Process Data

PostProcessData (PostProcessData)



Filtering

Opaque Layer Mask

Mixed...



Transparent Layer Mask

Mixed...



Shadows

Transparent Receive Shadow ☒

Overrides

Stencil



Renderer Features

▼ ☒ New Stars Render Feature (Stars Render Feature) ⋮

Name

NewStarsRenderFeature

▼ ☒ New Skybox Clouds Feature (Skybox Clouds Feature) ⋮

Name

NewSkyboxCloudsFeature

▼ ☒ New Depth Fog Render Feature (Depth Fog Render Feature) ⋮

Name

NewDepthFogRenderFeature

Add Renderer Feature

Additional Notes

- When first setting up the Skybox, you may need to reload the scene by entering and exiting playmode once.
- When setting up Stars configuration, you may need to manually refresh the Stars VFX by entering and exiting playmode once.
- Make sure you turn on Post-Processing effects, including Bloom. Bloom is critical for the sun to “pop”.

No bloom (left) vs bloom (right)



Credit for the Moon Albedo map to NASA's Scientific Visualization Studio.

Credit for the Moon Normal map to both NASA's Scientific Visualization Studio and Christian Petry.

Contact

If you encounter any issues at all, please don't hesitate to contact me at occasoftware@gmail.com